

Sri Sivasubramaniya Nadar College of Engineering, Chennai

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M.Tech (Integrated) Computer Science & Engineering Semester V

ICS1512 – Machine Learning Algorithms Laboratory

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Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering ¹
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1 Aim and Objective

1.1 Objective

To implement and analyze the performance of three clustering algorithms on the Human Activity Recognition (HAR) Using Smartphones dataset:

- **Model A:** K-Means clustering (with the Elbow method and Silhouette analysis to choose k).
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC) with Ward's linkage.

The clusters formed by each algorithm are visualized in two dimensions, compared with the ground-truth activity labels, and evaluated using internal and external clustering metrics.

1.2 Problem Statement

Input: 10,299 smartphone sensor windows, each described by 561 time- and frequency-domain features (accelerometer and gyroscope signals).

Output: A partition of the windows into clusters, without using the activity labels during training.

Goal: Determine how well unsupervised algorithms recover the six human activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) and which algorithm produces the most meaningful clusters.

2 Dataset Description

¹GitHub Repository: <https://github.com/username/repository>

Dataset Name	Human Activity Recognition Using Smartphones (UCI HAR)
Dataset Source	UCI Machine Learning Repository [1]
Number of Samples	10,299 windows (2.56 s each, 128 readings per window, 50 Hz sensors)
Number of Features	561 numeric features (plus the <code>Activity</code> label and the <code>subject ID</code>)
Number of Classes	6 (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING)
Subjects	30 volunteers (aged 19–48)
Missing Values	0
Duplicate Rows	0
Train-Test Split	Not applicable (unsupervised). The official train and test files were concatenated into one dataset (<code>har_combined.csv</code>) and all 10,299 windows were clustered.

2.1 Exploratory Data Analysis

The EDA covered the dataset overview, data types, missing-value and duplicate checks, descriptive statistics, class distribution, feature distributions, and a correlation matrix. Because there are 561 numeric features, distribution plots were limited to the first 30 features and the correlation heatmap to the first 40 features.

- **Overview:** shape (10299, 563); 561 `float64` feature columns, one integer column (`subject`) and one string column (`Activity`); memory usage about 44.2 MB.
- **Missing values:** none before or after cleaning; no duplicate rows.
- **Statistical summary:** all features are normalized by the dataset authors to $[-1, 1]$. For example, `tBodyAcc-mean()-X` has mean 0.274 and standard deviation 0.068, while `tBodyAcc-std()-X` has mean -0.608 , median -0.943 and 75th percentile -0.250 .
- **Classes:** `Activity` has 6 unique values; LAYING is the most frequent (1944 windows). The 6 classes are roughly balanced.

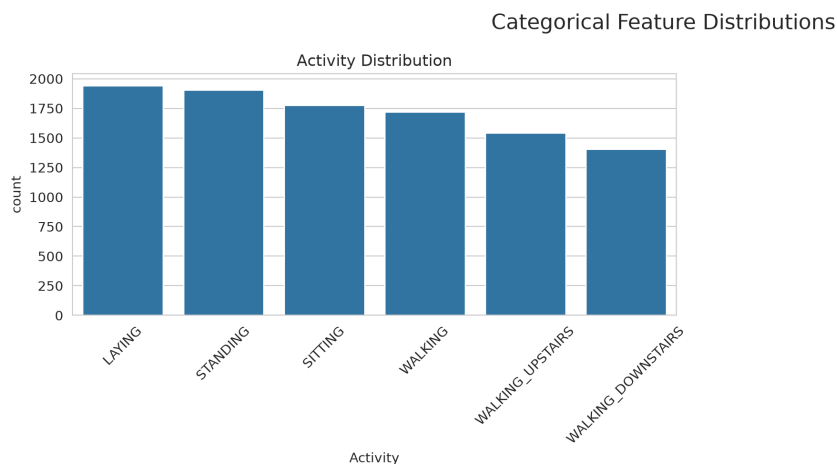
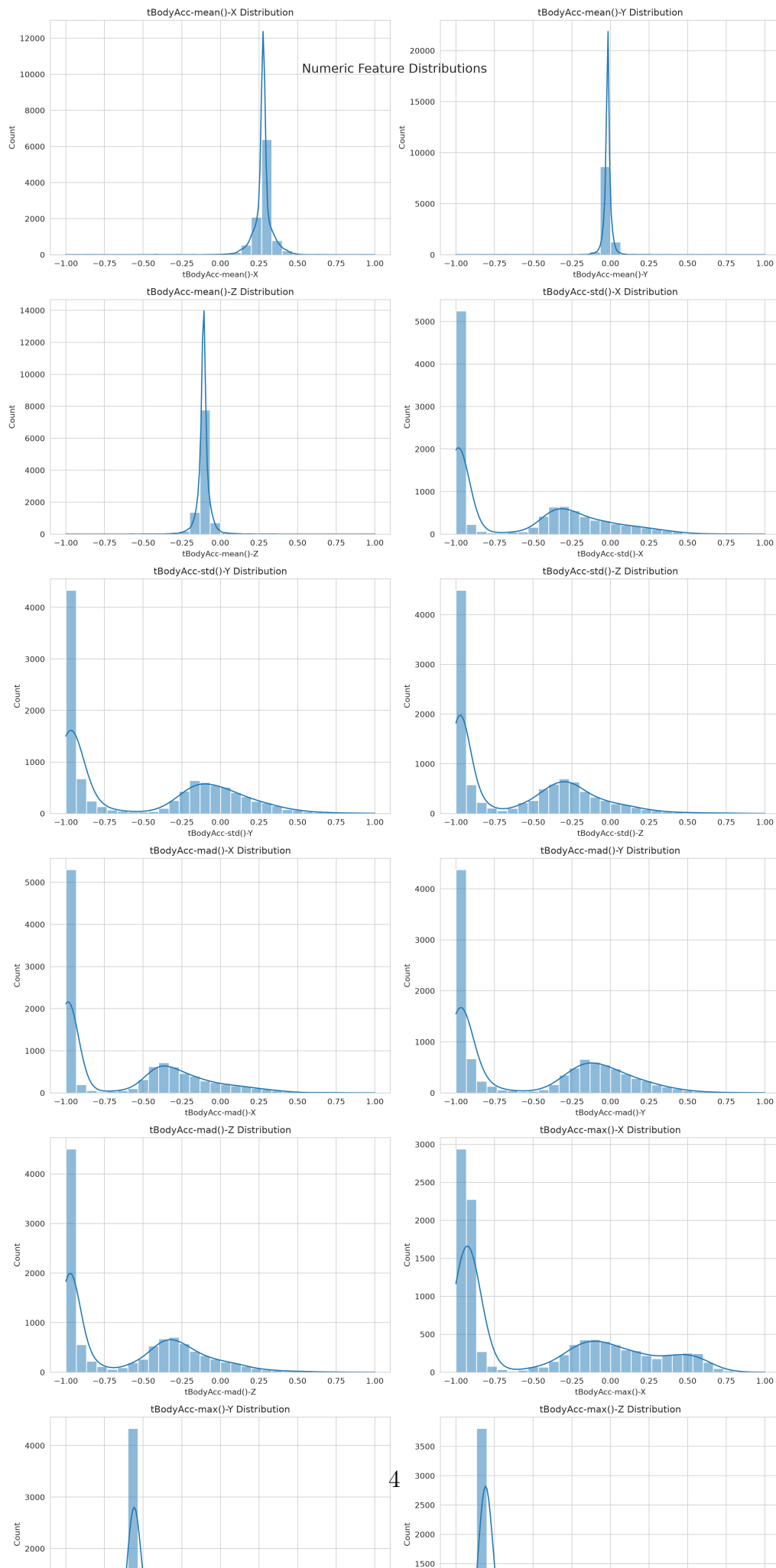


Figure 1: Class (activity) distribution.

Inference: The six activities have comparable numbers of windows (LAYING is the largest with 1944), so the dataset is close to balanced. No single class dominates, which means the external metrics (ARI, NMI) are not biased toward one large activity. This also confirms that the ground-truth labels are a fair reference for judging the clusters.



Inference: The features are bounded in $[-1, 1]$ and several are strongly skewed; the `std` features, for instance, have most of their mass near -1 with a long tail toward higher values, which is consistent with a median of -0.943 against a mean of -0.608 . Low-motion activities (sitting, standing, laying) produce values concentrated near the lower bound, while dynamic activities produce the tail. The multimodal, non-Gaussian shapes suggest that the data is not made of well-separated spherical blobs.

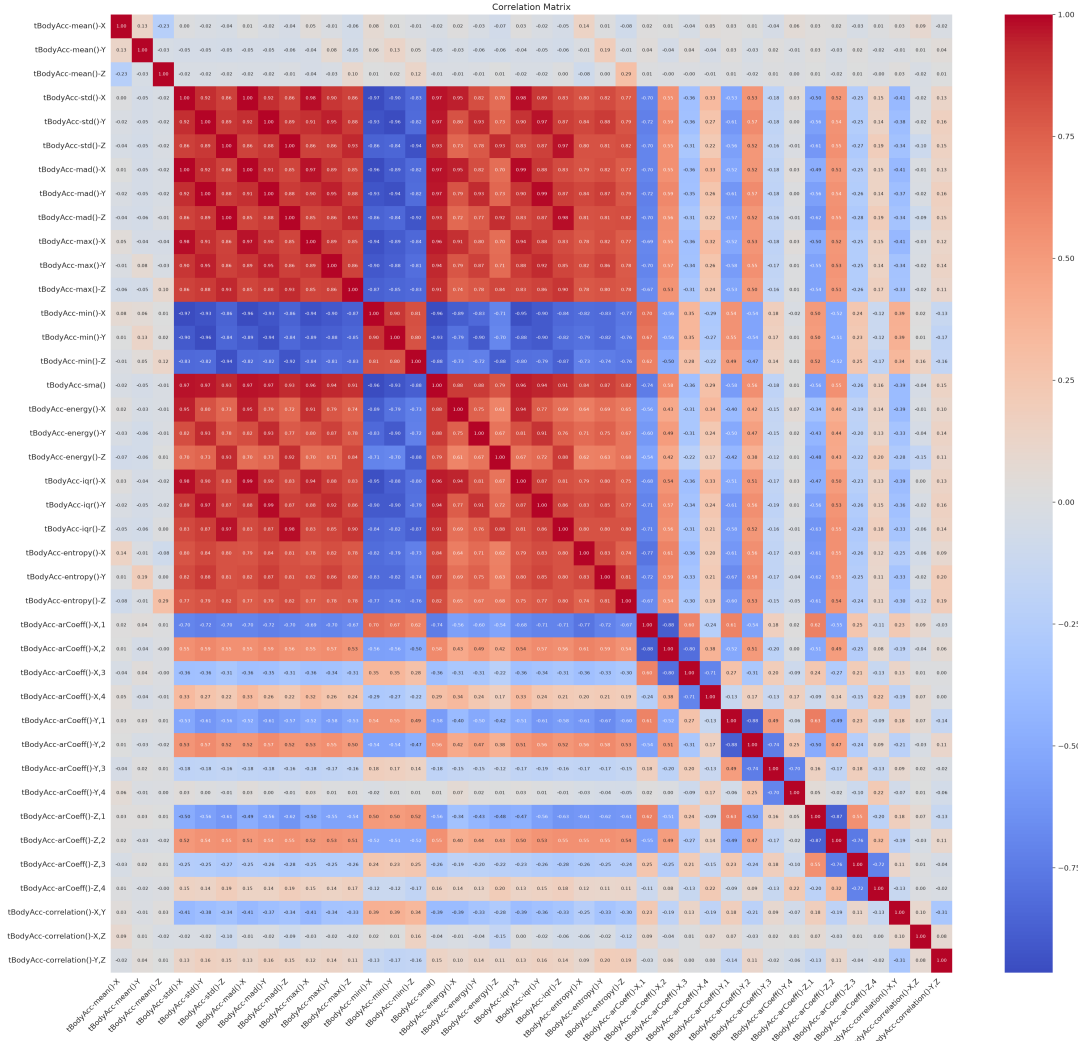


Figure 3: Correlation matrix of the first 40 numeric features.

Inference: The first 40 features are statistics of the same body-acceleration signal (mean, std, mad, max, min, energy, iqr, entropy, ...), and they show blocks of strong positive and negative correlation. This redundancy is expected for hand-crafted signal features. It indicates that the 561 dimensions have a much lower effective dimensionality, which motivates PCA for visualization and for DBSCAN.

3 Preprocessing Steps

1. **Loading and merging.** The raw files (`features.txt`, `X_*.txt`, `y_*.txt`, `subject_*.txt`) of the train and test folders were combined into one table. Feature names in

`features.txt` are not unique, so duplicates were made unique by appending a running suffix (`_2`, `_3`). The numeric activity IDs were mapped to their names using `activity_labels.txt`.

2. **Missing values.** Null-like strings (e.g. NA, ?, -) were converted to NaN and rows with NaN were dropped. No values were missing.
3. **Duplicates.** Duplicate rows were removed; none were found.
4. **Outliers.** IQR-based outlier removal was deliberately *skipped*: it combines the mask of all 561 columns with a logical AND and would discard almost every row.
5. **Encoding.** The `Activity` label was encoded with `LabelEncoder` (0–5). It was used only for the external evaluation and never as a model input.
6. **Feature selection.** The `Activity` and `subject` columns were removed, leaving a 10299×561 feature matrix.
7. **Scaling.** Features were standardized with `StandardScaler` (zero mean, unit variance) because K-Means, DBSCAN and Ward linkage are all distance-based.
8. **Dimensionality reduction.** PCA to 2 components was used to visualize the clusters. Clustering itself was performed in the full 561-dimensional standardized space.
9. **Train-test split / feature engineering.** Not applicable. The features were engineered by the dataset authors from the raw signals.

4 Implementation Details

4.1 Machine Learning Algorithms

K-Means. K-Means [2] partitions the data into k clusters by minimizing the within-cluster sum of squares (WCSS, inertia):

$$J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2. \quad (1)$$

The algorithm initializes k centroids (k-means++), assigns each point to its nearest centroid, recomputes each centroid as the mean of its points, and repeats until convergence. *Advantages:* simple, fast, scales well. *Limitations:* requires k in advance, assumes compact, roughly spherical clusters of similar size, is sensitive to initialization and outliers, and is forced to assign every point to a cluster.

DBSCAN. For a radius ϵ and a minimum count *minPts*, the neighborhood of a point p is $N_\epsilon(p) = \{q : d(p, q) \leq \epsilon\}$. A point is a *core point* if $|N_\epsilon(p)| \geq \text{minPts}$, a *border point* if it lies in the neighborhood of a core point but is not core itself, and *noise* otherwise. Clusters are maximal sets of density-connected points [3]. *Advantages:* finds arbitrarily shaped clusters, does not need k , and labels outliers as noise. *Limitations:* very sensitive to ϵ and *minPts*, struggles with clusters of varying density, and suffers in high-dimensional spaces where distances between points become nearly uniform.

Hierarchical Agglomerative Clustering (Ward). HAC starts with every point as its own cluster and repeatedly merges the pair of clusters that gives the smallest increase in total within-cluster variance [4]:

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2. \quad (2)$$

The merge history is displayed as a dendrogram and can be cut at any level to obtain a chosen number of clusters. *Advantages:* no random initialization, gives a full hierarchy, Ward tends to produce compact clusters. *Limitations:* $O(n^2)$ memory and time, merges are irreversible, and it is sensitive to the choice of linkage. Single linkage tends to chain, complete linkage favors compact clusters and is affected by outliers, average linkage is a compromise, and Ward behaves like K-Means.

4.2 Evaluation Metrics

- **Silhouette** [5]: $s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$, where $a(i)$ is the mean distance to points in the same cluster and $b(i)$ the mean distance to the nearest other cluster. Range $[-1, 1]$, higher is better.
- **Davies–Bouldin** [6]: average over clusters of the largest ratio of within-cluster scatter to between-centroid distance. Lower is better.
- **Calinski–Harabasz** [7]: $\frac{\text{tr}(B_k)/(k-1)}{\text{tr}(W_k)/(n-k)}$, the ratio of between- to within-cluster dispersion. Higher is better.
- **Adjusted Rand Index** [8]: chance-corrected agreement between the clusters and the true activities. 1 is a perfect match, about 0 is random.
- **Normalized Mutual Information:** $\text{NMI} = \frac{2 I(Y; C)}{H(Y) + H(C)} \in [0, 1]$, the shared information between true labels Y and clusters C .

Internal metrics use only the features; ARI and NMI use the activity labels purely for evaluation.

4.3 Experimental Setup

All models come from Scikit-Learn [9].

Python Version	
Scikit-Learn Version	
IDE	
Operating System	Linux
Hardware	

4.4 Hyperparameter Selection

Model	Parameter	Value
K-Means	Initialization / <code>n_init</code> / seed	k-means++ / 10 / 42
K-Means	Range of k searched	2 to 8
K-Means	Selected k (max. silhouette)	2
DBSCAN	ϵ	3.0
DBSCAN	<i>minPts</i> (<code>min_samples</code>)	10
DBSCAN	Distance	Euclidean
HAC	Linkage	Ward
HAC	Number of clusters	6 (number of true activities)
Dendrogram	Subsample / truncation	500 random windows / last 30 merges
Visualization	PCA components	2

4.5 Code Excerpts

The full implementation is in `exp8.py`. All figures and the metrics CSV are written to the folder `plots 8`.

```

1  # K-Means elbow + silhouette sweep
2  for k in k_range:
3      km = KMeans(n_clusters=k, init="k-means++", n_init=10, random_state
4                  =42)
5      labels = km.fit_predict(X)
6      wcss.append(km.inertia_)
7      sil_scores.append(silhouette_score(X, labels))
8
9  # DBSCAN
10 labels = DBSCAN(eps=3.0, min_samples=10).fit_predict(X)
11
12 # Hierarchical (Ward)
13 labels = AgglomerativeClustering(n_clusters=6, linkage="ward").
14     fit_predict(X)

```

5 Visualizations

5.1 K-Means: Elbow and Silhouette Analysis

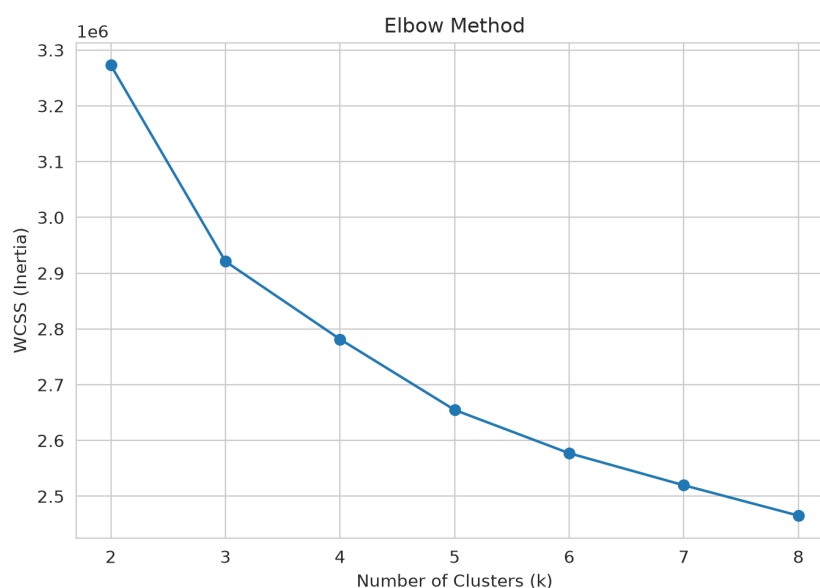


Figure 4: Elbow curve: number of clusters k versus WCSS.

Inference: WCSS decreases steadily from 3.27×10^6 at $k = 2$ to 2.47×10^6 at $k = 8$. The biggest single drop occurs from $k = 2$ to $k = 3$ (about 3.5×10^5), after which each additional cluster reduces WCSS by a much smaller and roughly constant amount. The curve therefore has only a weak elbow near $k = 3$, so the elbow method alone does not identify k decisively for this data.

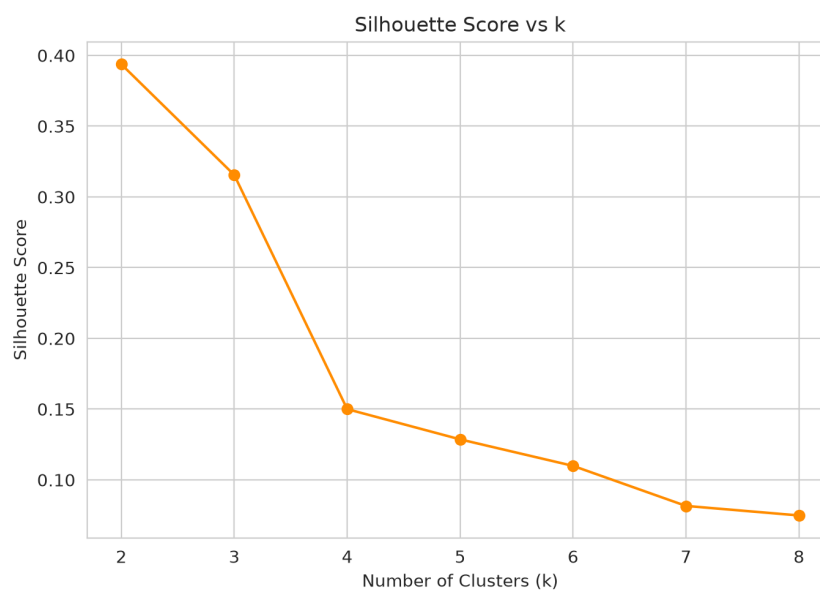


Figure 5: Silhouette score versus k .

Inference: The silhouette score is highest at $k = 2$ (0.394) and falls to 0.315 at $k = 3$,

then sharply to 0.150 at $k = 4$, and keeps decreasing to 0.075 at $k = 8$. This means the data has one clear coarse split, and every further subdivision produces overlapping clusters. The silhouette curve therefore selects $k = 2$.

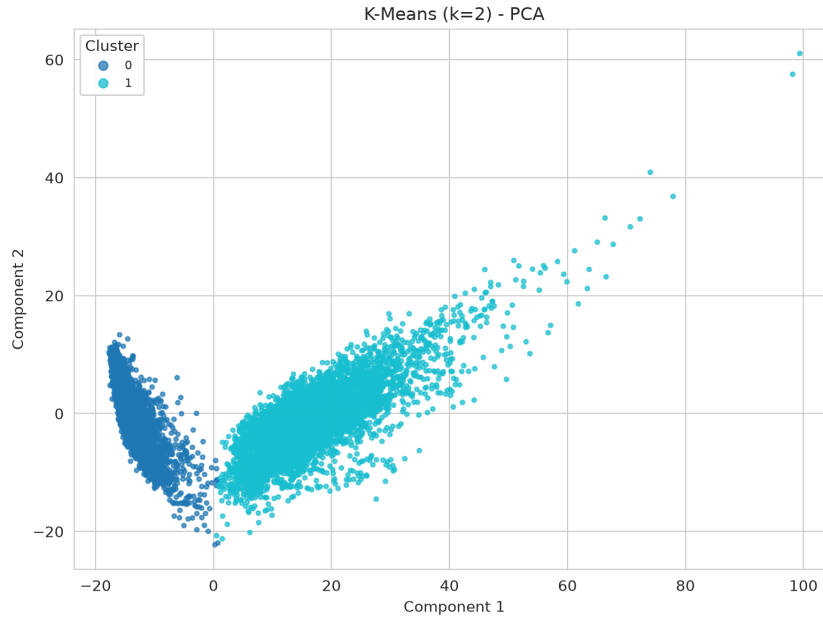


Figure 6: K-Means clusters ($k = 2$) projected on the first two principal components.

Inference: The two clusters occupy largely different regions of the PCA plane, which supports the high silhouette score of 0.394. For HAR data such a two-way split typically separates the static activities (sitting, standing, laying) from the dynamic ones (the three walking activities). Since the true data contains six activities, this partition is meaningful but coarse, which is reflected in the modest ARI (0.330).

5.2 DBSCAN

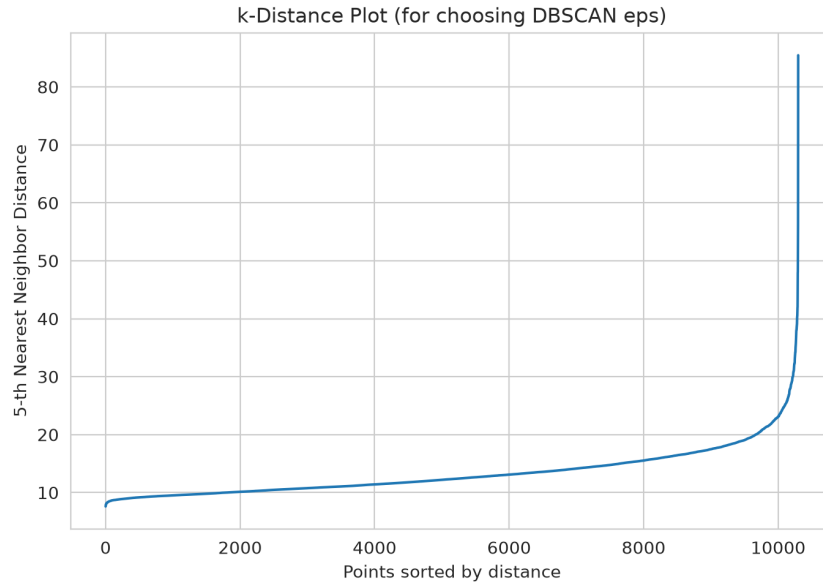


Figure 7: k-distance plot (sorted 5th-nearest-neighbor distance) used to guide the choice of ϵ .

Inference: The k-distance plot shows how far each point is from its nearest neighbors in the 561-dimensional standardized space. The knee of this curve is the usual choice for ϵ . In this space the neighbor distances are large compared with the value $\epsilon = 3.0$ that was used, which explains why no dense regions were found. Note that the plot used $k = 5$, while the DBSCAN run used $minPts = 10$.

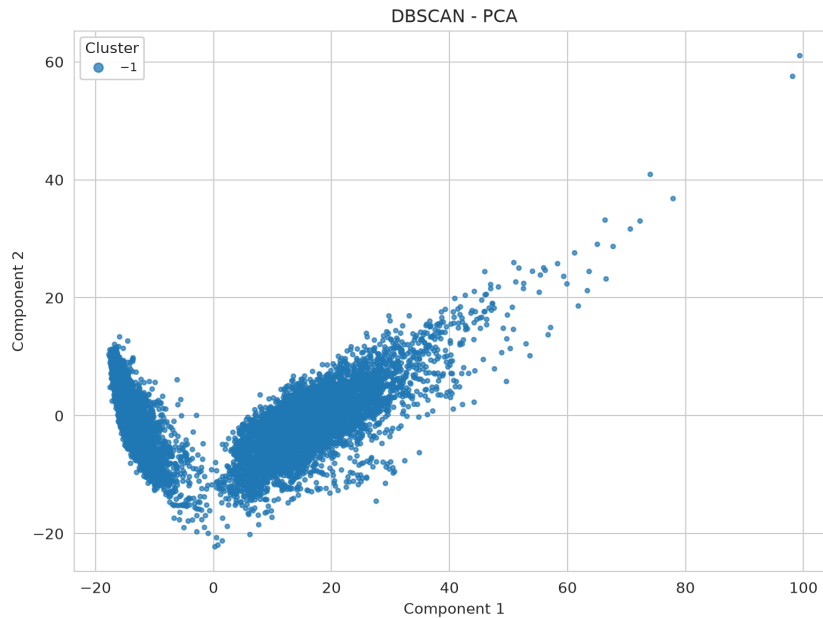


Figure 8: DBSCAN result on the PCA projection ($\epsilon = 3.0$, $minPts = 10$).

Inference: With $\epsilon = 3.0$ and $minPts = 10$ DBSCAN found 0 clusters and labeled all 10,299 points (100%) as noise, so the plot shows a single noise label. No point had at

least 10 neighbors within radius 3.0. This shows how strongly DBSCAN depends on ϵ in high-dimensional data, where distances concentrate and are large.

5.3 Hierarchical Clustering

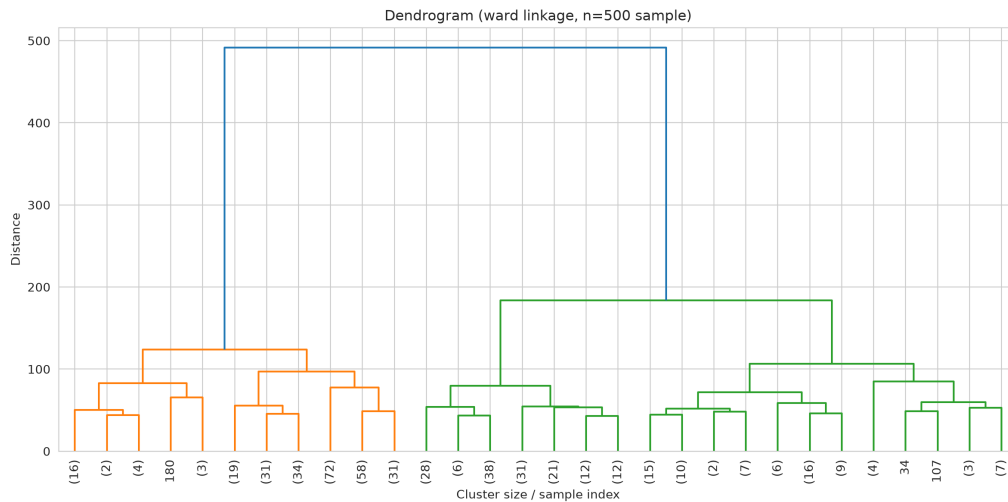


Figure 9: Dendrogram (Ward linkage, random sample of 500 windows, truncated to the last 30 merges).

Inference: The dendrogram is truncated to 30 leaves so that it remains readable. Ward linkage merges small groups at low heights and the last merges occur at much larger distances, which shows that a small number of well-separated groups exists at the top of the hierarchy. Cutting the tree at six clusters gives the same number of groups as the true activities.

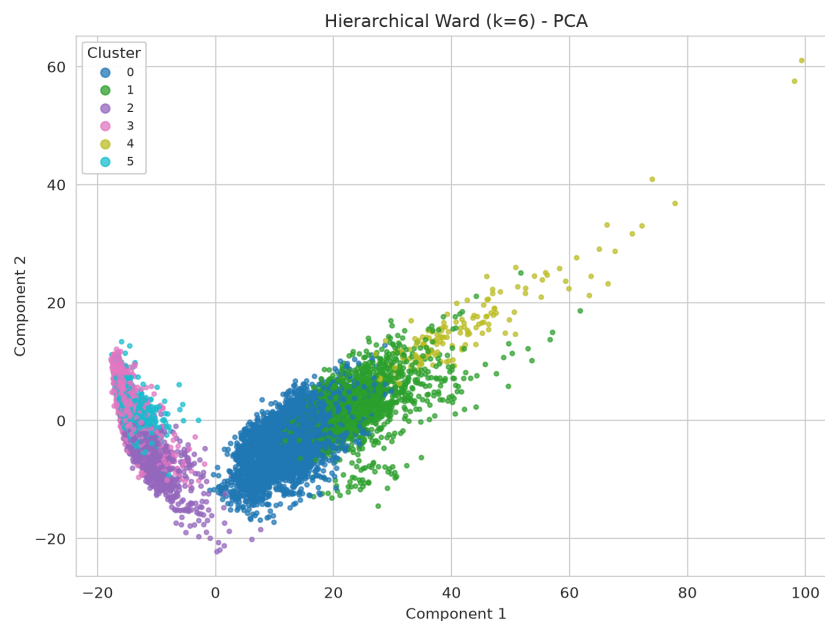


Figure 10: Hierarchical (Ward, 6 clusters) result on the PCA projection.

Inference: Cutting the hierarchy into six clusters produces groups that follow the activity structure more closely than the two-cluster K-Means solution: ARI increases to 0.460 and NMI to 0.601. Neighboring clusters overlap in the two PCA dimensions, in line with the low silhouette score (0.117), because activities such as sitting and standing and the three walking variants have similar features.

5.4 Comparison of Algorithms

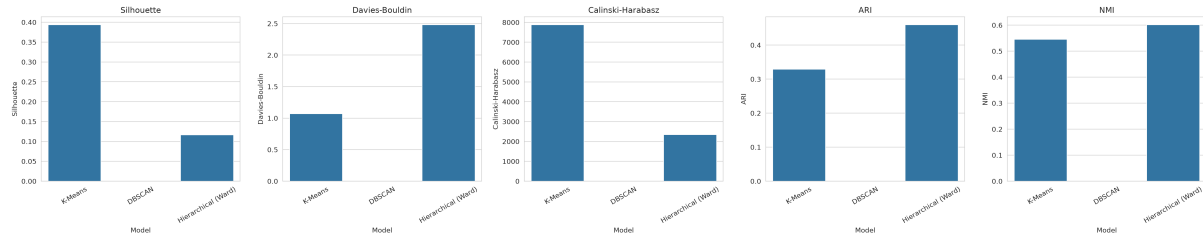


Figure 11: Comparison of Silhouette, Davies–Bouldin, Calinski–Harabasz, ARI and NMI across the algorithms.

Inference: K-Means ($k = 2$) is the best model on all three internal metrics (highest Silhouette and Calinski–Harabasz, lowest Davies–Bouldin), whereas Ward HAC ($k = 6$) is the best on the external metrics (ARI and NMI). DBSCAN has no bars for the internal metrics because it produced no valid clustering (metrics undefined), and its ARI and NMI are 0. The conflicting rankings are partly caused by the different numbers of clusters (2 versus 6) rather than by the algorithms alone.

6 Performance Tables

Table 2: K-Means elbow method and silhouette results.

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score	Δ WCSS from previous k
2	3,272,857	0.3937	—
3	2,921,078	0.3155	351,779
4	2,781,602	0.1500	139,476
5	2,654,790	0.1286	126,812
6	2,577,180	0.1099	77,610
7	2,519,774	0.0816	57,406
8	2,465,379	0.0749	54,395

Selected k : $k = 2$ was chosen by the maximum silhouette score (0.394). The elbow curve shows only a weak bend at $k = 3$, and both criteria agree that the strongest structure is a coarse one (static versus dynamic activities). For an activity-level (6-class) analysis, $k = 6$ is the natural alternative, but it has a much lower silhouette of 0.110.

Table 3: Clustering evaluation metrics on the HAR dataset.

Model	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NM
K-Means ($k = 2$)	0.3937	1.0707	7880.81	0.3296	0.545
DBSCAN ($\epsilon = 3.0$, $minPts = 10$)	–	–	–	0.0000	0.000
Hierarchical Ward ($k = 6$)	0.1170	2.4820	2349.67	0.4599	0.601

DBSCAN internal metrics are undefined (shown as “–”) because all points were labeled as noise and fewer than two clusters existed.

7 Overfitting and Underfitting Analysis

Clustering is unsupervised and there is no train–test split, so overfitting and underfitting are interpreted in terms of how finely the model partitions the data.

- **Underfitting (too few clusters).** K-Means with $k = 2$ has the best internal scores but merges six different activities into two groups; its agreement with the true labels is limited (ARI 0.330, NMI 0.545). This partition is too coarse to separate, for example, sitting from standing.
- **Overfitting (too many clusters).** WCSS decreases monotonically with k (from 3.27×10^6 to 2.47×10^6), exactly as training error decreases with model complexity, so it cannot pick k by itself. Silhouette acts as the validation signal: it falls from 0.394 to 0.075 as k grows, showing that additional clusters split otherwise continuous groups and lead to overlapping clusters.
- **Over-fragmentation in DBSCAN.** With $\epsilon = 3.0$ the neighborhoods are too small in the 561-dimensional space, so every point is isolated and treated as noise. This is an extreme case of overly local structure with no generalization at all.
- **Ward with $k = 6$.** Forcing six clusters improves the label agreement (ARI 0.460) but yields low silhouette (0.117) and a high Davies–Bouldin index (2.48), which indicates that some activities are inherently not separable in the feature space.

8 Bias–Variance Analysis

For clustering, bias corresponds to the strength of the assumptions a model makes about cluster shape, and variance to the sensitivity of the result to initialization, parameters and small changes in the data.

- **K-Means: high bias, low variance.** It assumes compact, spherical, similar-sized clusters, which the HAR classes do not satisfy, so its bias is high. Variance is low because of `k-means++` with 10 restarts and a fixed seed, although the result depends strongly on the chosen k (silhouette $0.394 \rightarrow 0.075$ over $k = 2\text{--}8$). Stability could be checked by re-running with different seeds and computing the ARI between runs.

- **DBSCAN: low shape bias, high variance.** It makes no assumption about cluster shape, but its output changes drastically with ϵ and *minPts*; in this experiment a single parameter setting produced no clusters at all.
- **Hierarchical (Ward): moderate bias, low variance.** It is deterministic, so there is no initialization variance. Ward's variance-minimizing criterion gives a compactness bias similar to K-Means, and greedy, irreversible merges can propagate early mistakes.
- **Effect of dimensionality.** With 561 correlated features, distances between points become less informative. This raises variance for density-based methods and biases all distance-based methods toward the dominant directions of variation.

9 Observations and Conclusion

9.1 Observations

1. **Which algorithm produced the most meaningful clusters?** Hierarchical Ward clustering ($k = 6$) matched the activity labels best (ARI 0.460, NMI 0.601). K-Means ($k = 2$) had the best internal quality (silhouette 0.394, Davies–Bouldin 1.07, Calinski–Harabasz 7880.8) and captured the coarse static versus dynamic structure. DBSCAN produced no usable clusters. The two leading models used different k , so the comparison is not exactly like-for-like.
2. **How sensitive was K-Means to k ?** Very sensitive. The silhouette dropped from 0.394 ($k = 2$) to 0.150 ($k = 4$) and 0.075 ($k = 8$), and the WCSS curve had no sharp elbow (the largest reduction was 351,779 from $k = 2$ to 3, then smaller, nearly constant reductions).
3. **Did DBSCAN detect noise or small clusters effectively?** No. For $\epsilon = 3.0$ and *minPts* = 10 all 10,299 points became noise. In a 561-dimensional standardized space, typical distances between points are much larger than 3.0, so no point has enough neighbors. Reducing the dimensionality (e.g. PCA) and choosing ϵ from the knee of the k -distance plot would be required.
4. **How does linkage choice affect hierarchical clustering?** Only Ward linkage was run in this experiment. In general, single linkage merges through nearest points and tends to chain and to form one giant cluster; complete linkage produces compact clusters but is sensitive to outliers; average linkage is a compromise; Ward minimizes the increase in variance and gives balanced, compact clusters similar to K-Means, which suits standardized, roughly continuous data such as HAR.
5. **Which internal metric best matched visual intuition?** Silhouette (together with Calinski–Harabasz and Davies–Bouldin, which ranked the models in the same order) agreed with the PCA scatter plots in favoring the clean two-group K-Means result over the overlapping six-group Ward result. However, none of the internal metrics agreed with ARI/NMI, which favored the six-cluster solution. Internal metrics reward compactness and separation, not agreement with the six real activities.

9.2 Discussion

Strengths. The pipeline is fully automated (loading, EDA, scaling, three algorithms, five metrics, saved figures). Using the labels only for evaluation keeps the study unsupervised.

Limitations. K-Means and Ward were compared at different values of k ; DBSCAN was not tuned successfully; only Ward linkage was tested; and clustering was done in the full 561-dimensional space, where distances are less reliable. **Possible improvements.** Also report K-Means at $k = 6$; apply PCA (e.g. 95% variance) before DBSCAN and select ϵ from the k-distance knee with *minPts* matching the plot; compare single, complete, average and Ward linkage; add t-SNE visualizations; and replace the confusion matrix by a contingency table (cluster versus activity) to show which activities are merged.

9.3 Conclusion

Three clustering algorithms were applied to the UCI HAR dataset (10,299 windows, 561 standardized features). K-Means with the silhouette-selected $k = 2$ gave the best internal scores (silhouette 0.394) and reflected the split between static and dynamic activities. Hierarchical Ward clustering with $k = 6$ agreed best with the true activities (ARI 0.460, NMI 0.601) but with weaker cluster separation (silhouette 0.117). DBSCAN with $\epsilon = 3.0$ and *minPts* = 10 failed, labeling every point as noise, which demonstrates its sensitivity to parameters in high-dimensional spaces. Overall, the HAR activities form overlapping regions in feature space, so unsupervised clustering recovers the coarse structure well but does not fully separate the six activities.

References

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